

Course-Plan

Spring Semester, 2025

School	ENGINEERING
Department	Computer Science and Engineering
Course Code	CSMT 501
Course Name	Advanced Algorithms and Data Structures/Algorithms & Complexity
Instructor	Arindam Karmakar

1. Abstract

This course teaches techniques for the design and analysis of efficient algorithms, emphasizing methods useful in practice. The main focus is on techniques for constructing correct and efficient algorithms, and on tools to reason about them. Design paradigms include greed, divide-and-conquer, dynamic programming, reduction to network flow, and the use of randomness. A second focus point is computational intractability. NP-complete problems are covered, as well as ways to deal with them. The course forms a foundation for all areas of computer science.

2. Objective

- (a) Problem-solving skills and algorithm design.
- (b) Analyze the asymptotic performance of algorithms.
- (c) Write rigorous correctness proofs for algorithms.
- (d) Demonstrate a familiarity with major algorithms and data structures.
- (e) Apply important algorithmic design paradigms and methods of analysis.
- (f) Design efficient algorithms in common engineering applications.

3. Prerequisites of the Course

- (a) Familiarity with Data structures and Discrete mathematics
- (b) Good knowledge in programming language like C or C++

4. Course-Plan

Part 1: Introduction to algorithms	
Introduction to Algorithms, examples of simple algorithms, relevant discrete math regarding functions, sums, sets, counting, probability, matrices, induction, correctness of Algorithms	5
Running time, efficiency, tractable and intractable problems Asymptotic order of growth, O , Ω , and Θ notation	3
Part 2: Fundamental Algorithms	
Brute force & exhaustive search algorithms, Sorting, Depth-First Search, Breadth-First Search, Directed Acyclic Graphs, Topological Sorting, Strongly-Connected Components	6
Part 3: Algorithm Paradigms	
Divide & Conquer Algorithms I: Recurrence Relations, The Master Theorem, Applications of the Master Theorem Binary Search, Merge sort, Counting Inversion, Quick sort, Integer Multiplication	5
Divide & Conquer Algorithms II: Closest pair problems, The Median-of-Medians Algorithm, Strassen's algorithm for matrix multiplication	5
Greedy Algorithms: Coin Exchange, Interval Scheduling Shortest Paths, Dijkstra's Algorithm, Minimum Spanning Trees, Prim's and Kruskal's Algorithm	5
Dynamic Programming: Memoization, Optimal substructure, Weighted Interval Scheduling, Knapsack Problem, Edit distance, Bellman-Ford Algorithm, Floyd-Warshall Algorithm	5
Part 4: Advanced Topics	
Polynomial time algorithms, P and NP class, NP-hardness, NP-completeness NP-Reductions	3
Network flow, Bipartite matching, Introduction to Randomized & Approximation Algorithms	3

5. Suggested Reading:

Textbook

1. Dasgupta, Sanjoy, Christos Papadimitriou, and Umesh Vazirani. Algorithms. McGraw-Hill
2. Kleinberg, Jon, and Eva Tardos. *Algorithm Design*. Addison-Wesley

Supplementary Texts

1. T. H. Corman, C. E. Leiserson, R. L. Rivest, and C. Stein: Introduction to Algorithms. 3rd Edition. MIT Press
2. Data Structures and Algorithms by Alfred V. Aho, John E. Hopcroft, Jeffrey D. Ullman

6. Student Evaluation

Sl.no	Component	Marks
1	Mid-term	30
2	Test I :	10
3	End-Term	50

Total : 090

6. Pedagogy

Teaching-Learning Methods to be used

- Lectures and Discussion
- Presentations
- Laboratory
- Quiz

7. Expected Outcome:

1) On completion of this course you should have gained an understanding of algorithmic / data structure language and notation, including order notation, and how to calculate the running times of algorithms. You should also understand how to estimate, profile, and measure algorithm complexity and performance.

Upon completion of the course the student should have the following learning outcomes. They will be able to:

	Course Outcomes	Bloom's Taxonomy Level
CO1	Define, compare, analyse, and solve general algorithmic problem types: sorting, searching, string processing, graphs, and geometric algorithms	Understand
CO2	Analyze worst-case running times of algorithms using asymptotic analysis..	Understand
CO3	.Apply the key algorithmic design paradigms: brute force, divide and conquer, decrease and conquer, transform and conquer, greedy, dynamic.	Apply
CO4	Implement, empirically compare, and apply fundamental algorithms and data structures to real-world problems.	Analyze
CO5	can recognize new problems that are solvable using the techniques learned in this course, and design new algorithms for similar problems	Create